Akunna Onyekachi

Wake Forest, NC • 919-780-1919 • akunna1mail@gmail.com

Portfolio • GitHub • LinkedIn

SUMMARY

GovTech Geospatial Developer with 8 years of experience in GIS, data analytics, application development, workflow automation, and Tier 3 technical support, specializing in Python, R, JavaScript/TypeScript frameworks and libraries, Microsoft Power Platform, SQL, and Esri interactive mapping solutions. Adept at delivering data-driven solutions to improve public-sector operational decision-making and service delivery.

CAREER GOALS

Passionate about advancing geospatial and IT solutions through expertise in Esri ArcGIS, TypeScript, and data analytics tools. Seeking exciting new opportunities to apply technical skills, solve real-world challenges, and contribute to:

1. Designing GIS tools for spatial analysis, infrastructure planning, demographic forecasting, and public safety
2. Building apps integrating APIs, data analytics, databases, and visualization tools to support resource management and program impact

EDUCATION

The University of North Carolina at Chapel Hill, NC — Aug 2019 – Dec 2023

- **Majors:** M.S. in Information Science (Web/Mobile App Development and Data Science Focus) and B.S. in Environmental Science (Geographic Information Systems Focus)
- **Coursework:** APIs, CI/CD, Cloud Computing, Data Modeling, Geospatial Statistics and Analysis, Remote Sensing, SQL/NoSQL Databases, Version Control, Web Development and Optimization

TECHNICAL SKILLS

- **App Development & Platforms:** MS Teams, Next.js, Node.js, React.js, SharePoint, Tailwind CSS, TypeScript
- **Cloud & Databases:** Firebase, Google Cloud, IBM MaaS360, MongoDB, MySQL, Oracle, PostgreSQL
- **Data Science & GIS:** ArcGIS, AutoCAD, GEE, Leaflet.js, Mapbox, Power BI, Python, QGIS, R

WORK EXPERIENCE

IT Analyst, City of Durham, Durham, NC — Oct 2025 – Present

- Support Fire Department IT systems using GIS and data analysis software to automate administrative processes and improve efficiency, including Oracle Database integration, mobile device management, and support for Computer Aided Dispatch (CAD) systems
- Build and improve data workflows using Python and Power BI to clean, analyze, and visualize staffing, operational, and financial data, improving data accuracy and supporting decision-making
- Develop internal applications and automated workflows using Next.js and Microsoft Power Platform (Power Apps, Power Automate), integrating SharePoint, TeleStaff, Active911, ESO Suite, and Supabase to streamline administrative processes and data handling

Geospatial Developer, Sambus Geospatial, Remote — June 2025 – Present

- Design and implement digital parcel mapping solutions using ArcGIS, digitizing over 3,500 parcels across major developing African cities such as Lagos and Malabo, preventing duplicate land sales and improving property verification
- Build prototype real estate information systems with ArcGIS GDB, integrating zoning, land use, and cadastral data into a centralized database to strengthen regional growth initiatives, enabling planners, realtors, and government agencies to validate land records
- Collaborate with regional agencies and private developers to improve property accessibility, using drone and satellite images and AutoCAD to georeference 220+ site plans, convert them to shapefiles, and address missing or fragmented land records

Full Stack Developer, FaithTech, Remote — Aug 2024 – Aug 2025

- Engineered a scalable GIS-enabled admin platform for ShareBibles' distribution using Firebase, Mapbox GL JS, Next.js, and ShadCN with Tailwind CSS for UI styling, and Git for version control, supporting global ministry operations
- Integrated PostgreSQL and Drizzle ORM for real-time data tracking and analysis, enabling over 400 teams worldwide to optimize Bible sharing strategies through seamless and efficient data querying

GIS Developer - Contract, DataWorks NC, Durham, NC — Apr 2024 – July 2024

- Analyzed high-value commercial property taxes, identifying over $50 million in uncollected revenue, focusing on high taxes in gentrifying neighborhoods with rising valuations impacting local residents
- Developed and enhanced a demographic mapping web application (i.e., the Durham Compass), utilizing Mapbox, Vue.js, and Node.js to visualize geospatial data from PostGIS and other sources, helping to comprehend the census and demographic makeup of Durham
- Designed and automated ETL pipelines using R and packages like tidycensus to extract, clean, and load census, real estate, property tax, and eviction data into the web application, streamlining demographic and housing metric integration

KEY GIS AND SOFTWARE PROJECTS

Akunna Writes (Built with: Firebase, React.js, Tailwind CSS, Vite.js)

- Developed a blog web app for posting short narratives in various languages, enabling users to log in, view, and contribute
- Purpose: To promote Igbo language literacy and cultivate a multicultural community by offering a platform for diverse voices to share stories and translations, while also serving as a source of personal motivation during challenging times

Durham Fire Compass (Built with: Firebase, Leaflet.js, Next.js, Python, R, Recharts, Tailwind CSS, Turf.js, TypeScript)

- Built cross-filtering dashboards visualizing coverage, station performance, and other operational metrics, alongside a CAD simulator implementing dispatch logic based on distance, station inventory, and response time to simulate emergency resource allocation
- Purpose: Enables what-if scenario testing for staffing and apparatus levels, allowing users to immediately evaluate how changes in demand and supply impact coverage, response capability, and operational readiness

Level of Traffic Stress Road Classification and Bike Lane Recommendations (Built with: ArcGIS Pro, R)

- Classified road network data of Chapel Hill, NC into LTS categories using lane count, number of intersections, and traffic flow, and created vector layers to map low-stress routes with recommended bike lane additions
- Purpose: To support safer cycling, assist UNC's Department of City and Regional Planning, and help reduce demand for limited campus parking

Wake County Public Transit Analysis (Built with: R (ggplot2, sf, tidytransit, tmap))

- Conducted a GIS-based analysis using GoRaleigh GTFS transit and U.S. Census socioeconomic data to identify neighborhoods in Wake County, NC with limited access to bus routes, applying 400-meter buffers around stops
- Purpose: To support fair transit planning by highlighting underserved areas and providing insights for local officials and planners

Zoning for Dummies (Built with: Leaflet.js, Next.js, R, Tailwind CSS, TypeScript)

- Developed an interactive, color-coded zoning web app for the Town of Wake Forest Planning Department, helping residents, students, and planners understand residential, commercial, and industrial zoning classifications
- Purpose: To make the zoning landscape more accessible and demystify urban planning concepts for the public

Additional information, links, and projects available on Portfolio and GitHub